2025 IEEE International Women in Engineering (WIE) Conference on Electrical and Computer Engineering (WIECON-ECE)
21–22 December 2025, Cox's Bazar, Bangladesh

Digitizing Conventional Water Meter Readings in Bangladesh Using a Scalable OCR-Based Extension Module

RAJIT PATIL ATRI*, MD INTESHAR ISHRAK*, RAJIA SULTANA*, LIZA AKTER*, ABIR AHMED†, MD. RIFAT HAZARI*, AND MD. SANIAT RAHMAN ZISHAN*
*Department of Electrical and Electronic Engineering, American International University-Bangladesh (AIUB), Bangladesh
†Department of Computer Engineering, American International University-Bangladesh (AIUB), Bangladesh
(20-43845-2, 20-44029-2, 20-44053-2, 20-42468-1)@student.aiub.edu, abir.ahmed@aiub.edu, rifat@aiub.edu, saniat@aiub.edu

***Abstract*—Manual meter reading in Bangladesh consumes significant man-month resources, and it makes the process both costly and inefficient. This study presents an image processing–based water meter reading system that automates data collection and reduces the dependence on manual inspections to address the issue. The framework integrates Optical Character Recognition (OCR) with Internet of Things (IoT) functionalities. A Raspberry Pi 3B+ and a high-resolution Pi Camera Module with Wi-Fi connectivity are used to capture and process images of analog water meters. The system was first validated through simulation in the PROTEUS Design Suite, and a prototype was implemented to check operational reliability under varying lighting and installation conditions. Experimental results showed that high-resolution imaging modules improved recognition performance by achieving up to 97% accuracy in controlled environments, while 90% in low-light conditions and 40% in dark settings. The proposed system provides a cost-effective and scalable pathway for modernizing utility management without costly infrastructure replacement by ensuring compatibility with existing analog meters widely deployed in Dhaka city.**

***Index Terms*—Image Processing, Internet of Things (IoT), Smart Utility Management.**

I. Introduction

The water supply and distribution system in Bangladesh faces persistent challenges due to water wastage, revenue leakage, and irregularities in traditional billing practices. Manual water meter reading is vulnerable to manipulation and inaccuracies that result in financial losses for utilities and increased costs for consumers. A report highlighted that nearly 20% of the water supplied daily by Dhaka Water Supply and Sewerage Authority (WASA) remains unaccounted. It is considered a substantial "system loss" in the distribution network [1]. Although WASA produces approximately 2.6 billion liters of water per day versus a demand of 2.5 billion liters, loss of nearly 520 million liters of water is reported each day for the inefficiencies and unmonitored usage. Such discrepancies emphasize the inadequacy of conventional practices and the urgent need for reliable, automated, and cost-effective water resource monitoring systems [2].

Recent advances in the IoT, embedded systems, and digital image processing have helped to progress in smart metering solutions. Smart water meters integrate sensing, communication, and computation. It provides accurate, real-time monitoring of water use, minimizing manual dependency and errors [3]. Methods using electronic flow sensors and wireless data transmission and image-based recognition with OCR technologies offer a cost-effective retrofit option for analog meters [4]. Faster-RCNN achieved 91.8% recognition accuracy on full meter images [5] while histogram equalization and template matching have shown feasibility in Bangladesh [6], [7]. OCR has also proven effective in related applications, such as smart parking systems, where it has been used successfully to recognize multilingual license plates under varying environmental conditions, further underscoring its robustness and broad applicability [8], [9].

Automatic meter reading (AMR) involves deep learning-based computer vision techniques developed to overcome challenges such as digit distortion, skew, poor lighting, and environmental noise. An end-to-end CNN framework combining YOLOv3 detection with orientation correction and CPD-guided digit regression achieved digit errors below 4% under challenging conditions [10]. Work on pointer meters introduced a text-guided detection strategy using Fast Oriented Text Spotting (FOTS), which achieved relative errors of less than 1% and demonstrated robustness under varying illumination and oblique viewing angles [11]. SSD-based detectors have also been applied for localizing digital regions in electronic meters with fine-tuned models attaining 100% detection accuracy on real-world test sets [12]. The UFPR-AMR benchmark dataset was introduced alongside a two-stage CNN framework that combined Fast-YOLO for region detection with CR-NET for digit recognition. It achieved 98.3% digit accuracy with real-time performance of 185 FPS [13]. More recently, a study addressed low-light conditions by integrating Mask-RCNN for skew correction, illumination enhancement, and ResNet101 regression, which reduced mean relative error to 2.2% [14]. Collectively, these studies demonstrate that CNN-based OCR methods outperform traditional handcrafted approaches, providing accurate, robust and scalable solutions for smart metering.

In addition to metering studies, a growing body of work has explored how IoT frameworks and automation can transform smart utility systems. An early contribution introduced a water metering model that combined ultrasonic sensors, M-Bus communication, and a cloud platform to provide both real-time and historical usage data [15]. IoT principles were applied to smart grids through the development of a peer-to-peer energy trading system for electric vehicle charging, which integrated M2M communication, MQTT, blockchain and machine learning and demonstrated resilience across simulations and hardware trials [16]. The scope of IoT has also been extended to electric vehicle safety, where a battery management system using Arduino and ESP8266 modules achieved over 93% measurement accuracy while enabling remote monitoring and mobile app integration [17]. More recently, an IR-based automation framework with Telegram bot integration demonstrated secure and low-cost device control which achieves reliable operation within ±10° and effective distances of 10–15 inches [18]. Overall, these studies highlight how IoT-driven connectivity, secure communication, and real-time analytics are shaping more sustainable and user-friendly smart utility ecosystems.

This paper presents the development of an OCR-based water meter reading system that applies image processing to automate water meter reading in Bangladesh. The framework integrates an RPi with a camera module and IoT connectivity to capture meter images, extract numerical readings using OCR, and transmit the data to a central database. The key contributions of this work are: (i) improving billing accuracy by reducing the dependency on manual inspections, (ii) ensuring compatibility with existing analog meters for a cost-effective deployment, and (iii) providing consumers with real-time access to usage data to promote transparency and accountability in resource management.

II. Methodology

This study formulates a framework by integrating an image processing system with IoT hardware. An RPi and a Pi Camera captures meter images, which are preprocessed and tested using OCR algorithms to extract numerical readings. The recognized data is transmitted via Wi-Fi or GSM to a cloud database for storage, analysis and billing.

Fig. 1 illustrates the proposed system architecture. The PiTunnel platform connects the Water Service Provider to the system for image acquisition and processing. The Pi Camera and an LED light regulated by a relay, captures meter images under consistent illumination. Images are processed by the OCR engine to extract numeric readings. The data are securely transmitted via Wi-Fi to the cloud with encryption to ensure integrity and confidentiality. This process reduces manual intervention, improves billing accuracy, and supports remote accessibility for efficient water resource management.

Fig. 2 shows the hardware setup of a system that consists of RPi, Pi Camera, LED light, and battery, which are integrated with a conventional water meter. The Pi Camera captures dial images under stable illumination from the LED while the RPi performs image acquisition and OCR recognition.

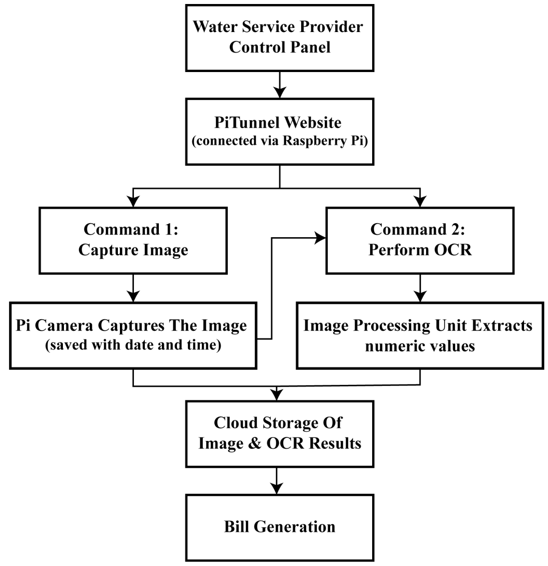

Fig. 1. Block Diagram of Image Processing-Based Meter Reading System.

It is powered by a battery for a standalone operation. This configuration demonstrates the system's practical applicability to automatically monitor water consumption.

Fig. 3 presents the workflow of the proposed system. It comprises three stages. At first, it extracts the Region of Interest (ROI) to isolate the digit display, then segments the ROI into individual digits and applies OCR to generate the final numerical reading. Each step of the process is explained in detail below:

A. Image Acquisition and Pre-processing

The dataset was collected using smartphone cameras with RGB images which is considered as an input. Images were converted into alternative color models and preprocessed through resizing and enhancement to reduce computational cost. All images were standardized to 512 × 512 pixels, ensuring consistency, accuracy, and reliable feature extraction under varying conditions.

B. Tilt Correction

Non-uniform camera positioning may introduce angular distortions that misalign digit regions and reduce recognition accuracy during image acquisition. To mitigate the problem, tilt correction is applied using the Hough Transform to detect

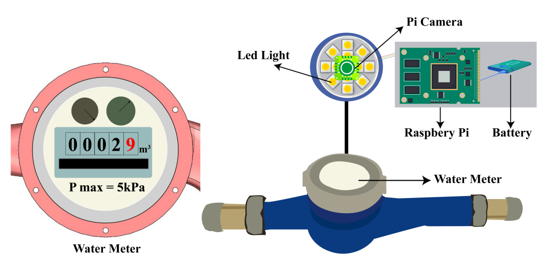

Fig. 2. Main setup for automated water meter reading.

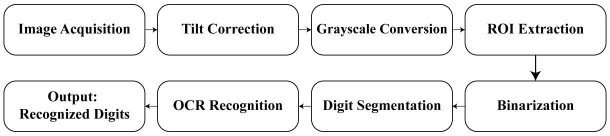

Fig. 3. Framework of the proposed automated water meter reading system.

skew angles. Then image rotated to restore proper alignment. The correction angle θ can be estimated as:

$$\theta = \arctan\left(\frac{\Delta a}{\Delta b}\right) \tag{1}$$

where Δa and Δb represent the displacement of the detected lines in the horizontal and vertical directions respectively. The image is rotated by $-\theta$ to obtain a properly aligned version for further processing.

C. Converting to Grayscale Image

In digital imaging, an RGB image consists of three color channels which yields over 16 million possible color combinations and consequently increasing computational complexity. Converting the image to grayscale reduces it to a single channel with 256 intensity levels that lowers processing cost while preserving essential structural information for recognition tasks.

$$I_{\text{gray}}(a,b) = 0.2989\,R(a,b) + 0.5870\,G(a,b) + 0.1140\,B(a,b) \tag{2}$$

where $R(a,b)$, $G(a,b)$, and $B(a,b)$ represent the red, green, and blue intensity components at pixel (a,b), respectively.

D. Extracting Region of Interest

In water meter images, the digit display occupies only a small portion of the frame while the background often introduces noise and redundancy. Processing the entire image increases computational cost and reduces accuracy. ROI extraction is applied to isolate the digit area for subsequent recognition. Mathematically, $I(a,b)$ represents the original image and the ROI can be defined as a submatrix of pixel intensities bounded by coordinates (a_1, b_1) and (a_2, b_2):

$$I_{\text{ROI}}(a,b) = \begin{cases} I(a,b), & \text{if } a_1 \le a \le a_2 \text{ and } b_1 \le b \le b_2 \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

where $I(a,b)$ denotes the pixel intensity of the original image at (a,b), and (a_1,b_1), (a_2,b_2) denote the top-left and bottom-right corners of the cropped region respectively. By restricting processing to this defined subregion, the system reduces noise, lowers computational cost and enhances the reliability of digit recognition.

E. Binarization

Intensity variations may still affect recognition after grayscale conversion and ROI extraction. Binarization converts the image into a binary form and enhance digit–background contrast. It enables more robust segmentation and recognition. The most common approach is global thresholding where a fixed threshold T is applied:

$$B(a,b) = \begin{cases} 1, & \text{if } I_{\text{gray}}(a,b) \ge T \\ 0, & \text{if } I_{\text{gray}}(a,b) < T \end{cases} \tag{4}$$

where $I_{\text{gray}}(a,b)$ is the grayscale intensity at pixel (a,b) and $B(a,b)$ is the resulting binary image. The system reduces noise and ensures that only the digit structures are retained for subsequent segmentation and OCR recognition by applying binarization.

F. Character Segmentation

Character segmentation separates individual digits from the binary image. It enables precise recognition by the OCR engine. This is typically achieved using connected component labeling, which identifies distinct pixel regions corresponding to each digit:

$$C_i = \{(a,b) \mid B(a,b) = 1,\ (a,b) \in \text{component } i\} \tag{5}$$

Here, C_i represents the set of pixels for the i^{th} digit. Segmentation isolates each digit to prevent overlap or fragmentation, improving the reliability and accuracy of OCR-based meter reading.

G. OCR Digit Recognition

In the final stage, the segmented digits are recognized using an OCR algorithm which classifies each digit into its corresponding numerical label. The recognition process can be expressed as a classification function:

$$d = \arg_{k \in \{0,1,2,\ldots,9\}} \max P(k \mid C_i) \tag{6}$$

where C_i is the i^{th} segmented digit, and $P(k \mid C_i)$ represents the probability that the digit belongs to class k. The output is the complete numerical reading of the water meter.

In summary, the proposed methodology integrates sequential image processing steps of grayscale conversion, ROI extraction, binarization, character segmentation and OCR recognition. It transforms raw water meter images into accurate digital readings. This structured process reduces computational complexity and enhances digit clarity. It also ensures reliable recognition and forms the foundation for automated water meter monitoring.

III. System Architecture and Implementation

The proposed system integrates hardware and software for efficient water meter automation. An RPi 3B+ with a Pi Camera, LED, relay and battery manages image capture and illumination. Also, it handles pre-processing, OCR-based recognition, and secure cloud transmission. Functionality was first validated through simulation to refine the architecture, identify issues early, and ensure a smooth transition to hardware deployment.

Fig. 4 shows the simulation setup developed in Proteus with a virtual RPi interface. This stage validated functionality and refined the design that enabled early issue detection, improving

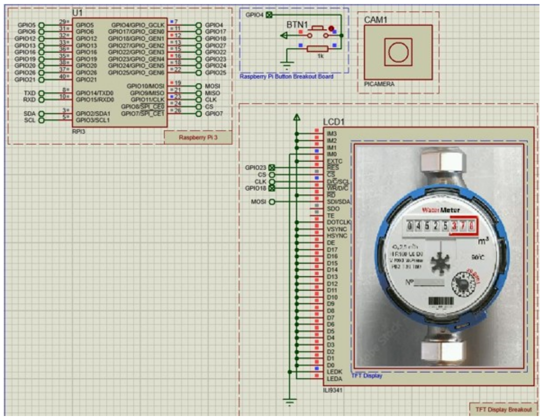

Fig. 4. Simulation Setup of Model.

reliability before hardware deployment. In the simulation, the RPi coordinated the Pi Camera, relay, and LED for virtual image capture and data flow verification that provides a solid foundation for the hardware prototype.

Fig. 5 shows the assembled hardware prototype where the Pi Camera captures high-resolution meter images under stable LED illumination with a relay synchronizing the process. This compact setup provides an efficient framework for accurate image acquisition and automated water consumption monitoring.

IV. Result Analysis

The proposed OCR-based water meter reading system was experimentally evaluated to assess its accuracy, efficiency, and reliability under varying conditions. Hardware tests, software performance, and data transmission outcomes are analyzed to validate the effectiveness of the design.

Fig. 6 presents the stepwise outputs of the OCR pipeline, beginning with raw input and concluding with the recognized meter readings. The workflow includes ROI extraction, grayscale conversion, binarization, and digit segmentation. Then OCR recognition used to generate the final numerical output, such as 00138. These pre-processing stages enhance image clarity and suppress noise and together it strengthen the reliability of digit recognition.

Table I summarizes the evaluation of different camera placements for water meter image acquisition. The Raspberry Pi Camera Module was tested at varying angles and distances to determine the most effective setup for accurate digit recognition. Results show that the close-up view in Sample 01 achieved the highest accuracy due to clear digit visibility. The long-shot view, shown in Sample 02, also delivered reliable recognition because the appropriate Region of Interest selection reduced the impact of reflections. In contrast, the angled placement which used in Sample 03 was affected by glare from external light. It introduced OCR errors even with the adjustments to the region of interest. The importance of proper alignment and illumination, with the strategically positioned LED enhancing clarity under low-light conditions, can improve overall recognition reliability.

Fig. 7 shows the accuracy test results. The close-up shot and the long-shot configuration achieved 100% recognition that confirms the effectiveness of proper camera positioning. The left-angle placement of the camera and LED produced only 20% accuracy; thus, it emphasizes the negative impact of glare and misalignment. Results indicate that positioning the camera directly above the meter with a slight offset from the LED provides balanced illumination. It minimizes reflections and ensures consistent recognition accuracy.

Table II highlights the effect of lighting conditions and ambient factors on image quality and OCR performance. Under good lighting, as shown in Sample 1, the system achieved the highest accuracy with clear images and reliable digit recognition. In low-light conditions, as shown in Sample 2, the model continued to perform strongly and demonstrated robustness against reduced illumination. The dark environment with minimal lighting used as Sample 3 resulted in OCR failure because poor visibility significantly distorted the digits. So, adequate lighting is essential during hardware installation to ensure consistent accuracy and dependable image processing across real-world scenarios.

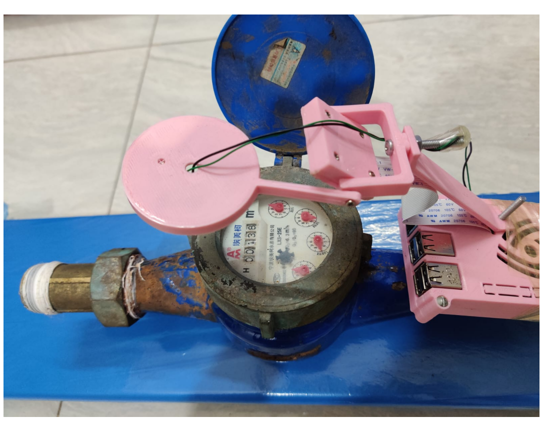

Fig. 5. Hardware Setup of Model.

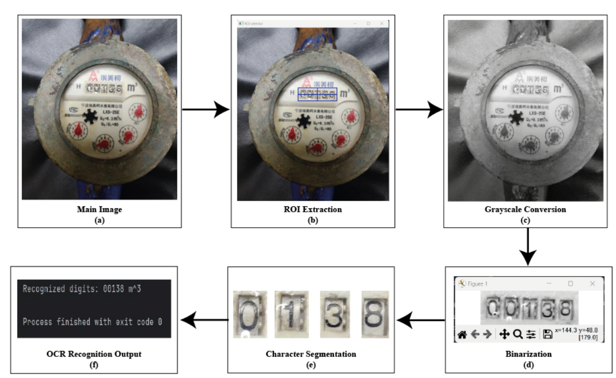

Fig. 6. Stepwise OCR processing workflow of the proposed system: (a) Original water meter image, (b) ROI extraction, (c) Grayscale conversion, (d) Binarization, (e) Character segmentation, and (f) Final OCR recognition output.

TABLE I
HARDWARE SAMPLE RESULT

Sample No.	Condition	Real Meter Image Index	Image Processing	OCR Predict Index	Result Status
1	Close Shot		Recognized digits: 00138 m^3 Process finished with exit code 0	00138	✓
2	Long Shot		Recognized digits: 00138 m^3 Process finished with exit code 0	00138	✓
3	Camera and LED Placed in Left Angle		Recognized digits: 01083 m^3 Process finished with exit code 0	01083	X

TABLE II
REAL SCENARIO RESULT

Sample No.	Condition	Real Meter Image Index	Image Processing	OCR Predict Index	Result Status
1	Good Lighting		Recognized digits: 16598 m^3	16598	✓
2	Low Lighting		Recognized digits: 01092 m^3	01092	✓
3	Dark place, Low Lighting		Recognized digits: 0019 m^3	0019	X

Fig. 7. Accuracy Test Result Ratio of Hardware Result.

Fig. 8 presents the digit recognition accuracy under different environmental conditions, ranging from 100% to 40%. Under good lighting, the system achieved 100% accuracy, while in low-light conditions, it maintained strong performance with 90% accuracy. But the performance dropped significantly to 40% in dark environments that simulated underground conditions, where low visibility and confined space limited recognition reliability. These results emphasize the critical importance of adapting to environmental factors in order to strengthen the robustness of OCR-based water meter reading systems.

Fig. 9 illustrates the software workflow of the proposed system, where Python processes images, applies OCR and generates automated invoices. The invoices are securely stored in Firebase, ensuring accurate billing and transparent, cloud-based access. To support remote environments, PiTunnel enables service providers to operate the RPi with only a device name and password. This integration of image processing, cloud storage and remote access enhances efficiency and reliability. The framework provides a scalable and user-friendly solution that fosters transparency and trust between providers and consumers.

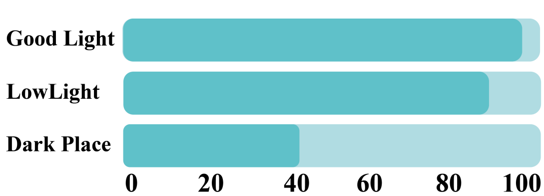

Fig. 8. Accuracy Test Result Ratio of Real Scenario Result.

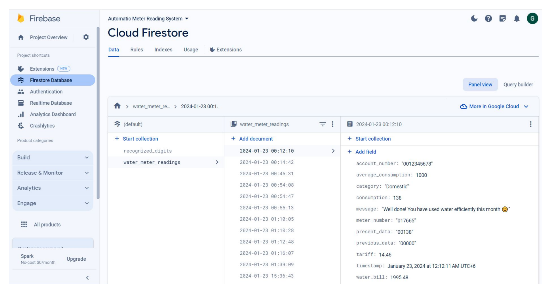

Fig. 9. Firebase Data Storage infrastructure.

Table III summarizes recent studies on automated digit recognition in metering and related domains. It highlights their methodologies, accuracy and limitations. While earlier approaches achieved competitive accuracy but many were affected by lighting, angular distortions, noise or limited flexibility across fonts and display types. The proposed RPi–based OCR method achieved accuracy (95–97%) with cost-effectiveness and compatibility with analog meters. It is a practical solution for deployment in resource-constrained settings.

This study introduces an OCR-based image processing framework for automated meter reading, achieving up to 97% accuracy under optimal lighting while maintaining 90% accuracy in low-light conditions and 40% in dark environments.

TABLE III
COMPARATIVE STUDY OF PROPOSED SYSTEM WITH PREVIOUS WORKS

Ref.	Study / Method	Approach	Accuracy	Key Limitation
[4]	Automated Number Recognition	Lightweight OCR (Moment Template Matching)	98.1%	Limited flexibility, harder with varied fonts
[6]	Image Processing Approach	Hough transform + ROI + template OCR	∼90% (varied)	Sensitive to lighting/tilt, limited scalability
[19]	Automated Digit Reading	Digit location algorithm + seven-segment classifier (HOG + neural network)	89–93% (depending on device)	Slight drop in performance due to segment detection errors
Proposed Work	RPi + OCR + IoT	Image capture (RPi + PiCam) + OCR + cloud integration	95–97% (depending on camera resolution)	Accuracy sensitive to lighting, but compatible with existing analog meters

Usually manual methods are time-consuming and error-prone. This system significantly reduces billing discrepancies and enhances efficiency. Its compatibility with existing analog meters avoids costly infrastructure replacement. It is a practical and scalable solution for improving billing accuracy, operational performance, and consumer trust in developing regions.

V. CONCLUSION

This study presents a practical solution for automating water meter data collection in Bangladesh using OCR based image processing. The system is implemented on a Raspberry Pi 3B+ with a high-resolution Pi Camera Module V2. It achieved up to 97% accuracy under optimal lighting, 100% in close-up and long-shot views but it declined to 20% with angled placement and 40% in dark settings. So the proposed solution shows cost-effective performance across varied conditions that ensures accurate readings while avoiding large-scale replacement of existing analog meters.

In future, the integration of dynamic LED illumination or advanced image enhancement and environmental sensing with deep learning techniques can improve recognition accuracy and robustness. This will improve billing precision, operational efficiency, and consumer trust. By maintaining compatibility with existing analog meters, the framework remains scalable and sustainable, supporting the wider adoption of smart utility management in developing regions like Bangladesh.

REFERENCES

[1] S. Rahman, "Dhaka wasa's incompetence leads to water tariff hike," *Prothom Alo*, February 2022, accessed:2025-09-01.

[2] J. Chowdhury and M. J. Islam, "Wasa suffers water bill loss, consumers made to pay," *The Business Standard*, April 2024, accessed: 2025-09-01.

[3] V. Nikić, D. Bortnik, M. Lukić, D. Vukobratović, and I. Mezei, "Lightweight digit recognition in smart metering system using narrow-band internet of things and federated learning," *Future Internet*, vol. 16, no. 11, pp. 1–24, OCT. 2024.

[4] X. J. Li and Y. Gao, "Digitisation of conventional water meters using automated number recognition," in *TENCON 2021 - 2021 IEEE Region 10 Conference (TENCON)*, Auckland, New Zealand, Dec 2021, pp. 470–475.

[5] L. Sun, Y. Yuan, S. Qiao, and R. Qi, "Detection of water meter digits based on improved faster r-cnn," *Journal of Computer and Communications*, vol. 12, no. 3, pp. 1–13, Jan 2024.

[6] S. Chaki, D. Shila, and N. Jahan, "Automation of water meter billing process based on digital image processing approach," *Journal of Advances in Technology and Engineering Research*, vol. 5, no. 5, pp. 207–218, Oct. 2019.

[7] T. Hassan, D. Das, S. I. Rhyme, S. Bal, M. R. Hazari, E. Jahan, C. A. Hossain, and M. A. Mannan, "Iot-based smart net energy meter with advanced billing feature for residential buildings including solar pv system," *International Journal of Power Electronics and Drive Systems (IJPEDS)*, vol. 15, no. 2, pp. 1254–1265, Jun. 2024.

[8] A. Himel, F. Sowmik, M. S. S. Oyon, M. Z. Islam, J. Bhowmik, and A. Ahmed, "Smart parking system with optical character reading technology," in *2024 27th International Conference on Computer and Information Technology (ICCIT)*, Bangladesh, Dec. 2024, pp. 1–7.

[9] F. Faisal, S. K. Das, A. H. Siddique, M. Hasan, S. Sabrin, C. A. Hossain, and Z. Tong, "Automated traffic detection system based on image processing," *AIUB Journal of Science and Engineering (AJSE)*, pp. 1–7, Jun. 2020.

[10] Q. Hong, Y. Ding, J. Lin, M. Wang, Q. Wei, X. Wang, and M. Zeng, "Image-based automatic watermeter reading under challenging environments," *Sensors*, vol. 21, no. 2, pp. 1–18, Jan. 2021.

[11] Z. Li, Y. Zhou, Q. Sheng, K. Chen, and J. Huang, "A high-robust automatic reading algorithm of pointer meters based on text detection," *Sensors*, vol. 20, no. 20, pp. 1–17, Oct. 2020.

[12] C.-M. Tsai, T. D. Shou, S.-C. Chen, and J.-W. Hsieh, "Use ssd to detect the digital region in electricity meter," in *2019 International Conference on Machine Learning and Cybernetics (ICMLC)*, Kobe, Japan, Jul. 2019, pp. 1–7.

[13] R. Laroca, V. Barroso, M. Diniz, G. Gonçalves, W. Schwartz, and D. Menotti, "Convolutional neural networks for automatic meter reading," *Journal of Electronic Imaging*, vol. 28, no. 01, p. 013023, Feb. 2019.

[14] X. Wu, X. Shi, Y. Jiang, and J. Gong, "A high-precision automatic pointer meter reading system in low-light environment," *Sensors*, vol. 21, no. 14, pp. 1–19, Jul. 2021.

[15] N. Cherukutota and S. Jadhav, "Architectural framework of smart water meter reading system in iot environment," in *2016 International Conference on Communication and Signal Processing (ICCSP)*, Melmaruvathur, India, Apr. 2016, pp. 791–794.

[16] A. Noor, M. S. Ratul, A. I. Ahmed, H. Hassain, and A. Ahmed, "An iot based smart grid: Peer-to-peer energy trading for electric vehicles using m2m communication technology," in *2023 3rd International Conference on Robotics, Electrical and Signal Processing Techniques (ICREST)*, Dhaka, Bangladesh, Jan. 2023, pp. 289–293.

[17] K. Insia, A. Ahmed, E. Jahan, S. Ahmad, S. Barua, I. Ali, and M. R. Hazari, "Design and analysis of iot-based battery management and monitoring system for electric vehicle," *AIUB Journal of Science and Engineering (AJSE)*, Aug 2023.

[18] Adnan, M. S. Sharar Oyon, A. H. Himel, K. Mredul, A. Ahmed, M. T. Ali, C. A. Hossainl, and M. A. Rahman, "Design and optimization of an ir-based automation system with telegram bot integration for remote device control," in *2025 4th International Conference on Robotics, Electrical and Signal Processing Techniques (ICREST)*, Dhaka, Bangladesh, Jan. 2025, pp. 275–279.

[19] E. Finnegan, M. Villarroel, C. Velardo, and L. Tarassenko, "Automated method for detecting and reading seven-segment digits from images of blood glucose metres and blood pressure monitors," *Journal of medical engineering & technology*, vol. 43, no. 6, pp. 1–15, Nov. 2019.